

Saam Haghighat-Grami

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EDUCATION

- **Kennesaw State University** *January 2022 – May 2022, January 2024 – Present*
Mechatronics Engineering, B.S. CGPA: 4.00
- **Marine Corps Communication Electronics School** *August 2022 – October 2023*
Possesses Active Security Clearance CGPA: 4.00
Ground Radio Repair School
- **Chattahoochee Technical College** *August 2020 – December 2021*
General Education (Undecided Major) CGPA: 3.80

RESEARCH EXPERIENCE

- **Research Interests**
 - Collaborative robotics, aerial-ground robotics, mobile robotics, Autonomous robotics
- **Uncertainty-Aware River Debris Trajectory Prediction** *June 2026 - August 2026*
Undergraduate Researcher, SURP – Dr. Matt Marshall
 - Developing a real-time edge-deployed system that detects, tracks, and predicts the trajectories of floating river debris.
 - Utilized neural networks to address the deficiency of linear Kalman-filter motion predictors in turbulent riverine conditions, implementing a class-conditioned LSPIV-LSTM-DNN architecture with a Gaussian uncertainty head to provide calibrated multi-step trajectory forecasts over YOLOv26 and ByteTrack detections on NVIDIA Jetson Orin Nano.
 - Tools & technologies used: PyTorch, YOLOv26, ByteTrack, LSTM-DNN, Gaussian NLL Loss, LSPIV, OpenCV, NVIDIA Jetson Orin Nano, TensorRT, Python, Raspberry Pi 5
- **A2G – UAV and UGV Collaborative Search and Rescue** *November 2025 - Present*
Robotic Software Architect
 - Designing a UGV-UAV collaborative search-and-rescue system: the UAV surveys the environment while the UGV clears obstacles blocking access to points of interest using an onboard manipulator.
 - Leading the machine learning and computer vision pipeline that estimates obstacle mass from sensor-fused SLAM point clouds to determine which obstacles the UGV's manipulator can safely move, alongside UAV flight motion.
 - Tools & technologies used: ROS2, NAV2, OpenCV, SLAM, UAV, UGV, RGB camera, LiDAR, Depth camera, Raspberry Pi 5, segmentation
- **SoilBus – Soil Monitoring Probes Network** *August 2025 - December 2025*
Lead software developer
 - Building a self-offloading soil-monitoring mesh network for farmers: probes auto-discover neighboring nodes and relay sensor data to any node without a central hub.
 - Using an LSTM to process incoming NPK, EC, pH, temperature, and moisture readings alongside light-level data across the network.
 - Validated communication and routing across 5 physical nodes, with performance holding up to a simulated 26 nodes before degradation; self-healing not yet implemented.
 - Tools & technologies used: Arduino IDE, C++, ESP32, ad hoc networking, DHT-22, 7-in-1 soil sensor, lux sensor, LSTM
- **DARB – Drone Arm Robotic Backpack (Personal Project)** *May 2025 - Present*
Inventor
 - Drone docking system using a robotic arm attached to a mobile human wearable platform
 - Custom drone utilizes computer vision to align with IR LEDs on wearble platform. Robotic arm retrieves drone in flight. The drone will be able to launch and release from the arm. Autonomously operated
 - Tools & technologies used: ArduPilot, ESP32, Arduino IDE, Open MV, Computer vision, Nema 23 Stepper motors with encoders, Electrical systems, Matek H743 WLITE, Tensor Flow, Onshape, N20

PROFESSIONAL EXPERIENCE

- **EOSYS Industrial and Manufacturing System Integrator** *May 2025 – August 2025*
COOP Marietta GA
 - Created Human Machine Interface display creator for P&IDs
 - Designed and Modified PLCs
- **United States Marine Corps Reserve** *August 2022 - Present*
Ground Radio Repair Specialist Marietta GA
 - Specialized in troubleshooting, maintaining, and operating USMC radio systems in time sensitive situations.

TECHNICAL SKILLS

Languages: C++, Python, MATLAB, PLC, Ladder Logic, Solidworks, Onshape

Developer Tools: Arduino IDE, VS Code, Linux, TIA portal

Frameworks: ROS2, Tensor Flow, OpenCV

Coursework: Machine Learning, Control Systems, Microcontrollers, PLCs, Robot Synthesis, Mobile Robotics

POSITIONS OF RESPONSIBILITY

- **Squad Leader of Marines:** 4th Combat Readiness Regiment USMC